

Foundations of Programming

Assignment No. : 06

PROBLEM STATEMENT: Write a C function to compute the factorial of a number with and without recursion.

OBJECTIVE:

To demonstrate two different approaches for calculating the factorial of a number: **iterative (non-recursive)** and **recursive**, showcasing the versatility of problem-solving methods in programming.

THEORY:

- 1) Explain the difference between the iterative and recursive approaches to calculating the factorial of a number.
- 2) What is the base case in the recursive function for computing factorials, and why is it necessary?
- 3) How does the program handle invalid input, such as negative numbers, and why is this important?

PLATFORM: Linux

INPUT: `#include <stdio.h>`

```
int factorial(int n) {  
    if (n == 0 || n == 1)  
        return 1;  
    else  
        return n * factorial(n - 1);  
}
```

```
int main() {  
    int num;  
  
    printf("Enter a number: ");  
    scanf("%d", &num);
```

```

    if (num < 0) {
        printf("Factorial is not defined for negative numbers.\n");
    } else {
        printf("Factorial using recursion: %d\n", factorial(num));
    }

    return 0;
}

```

OUTPUT:

Enter a number: 45

Factorial using recursion: 0

main.c	Run	Output
<pre> 1 #include <stdio.h> 2 3 int factorial(int n) { 4 if (n == 0 n == 1) 5 return 1; 6 else 7 return n * factorial(n - 1); 8 } 9 10 int main() { 11 int num; 12 13 printf("Enter a number: "); 14 scanf("%d", &num); 15 16 if (num < 0) { 17 printf("Factorial is not defined for negative numbers.\n"); 18 } else { 19 printf("Factorial using recursion: %d\n", factorial(num)); 20 } 21 22 return 0; 23 } 24 </pre>	Run	<pre> Enter a number: 45 Factorial using recursion: 0 === Code Execution Successful === </pre>

main.c	Output
<pre>1 #include <stdio.h> 2 3 int main() { 4 int num, fact = 1; 5 6 printf("Enter a number: "); 7 scanf("%d", &num); 8 9 if (num < 0) { 10 printf("Factorial is not defined for negative numbers.\n"); 11 } else { 12 for (int i = 1; i <= num; i++) { 13 fact = fact * i; 14 } 15 printf("Factorial without recursion: %d\n", fact); 16 } 17 18 return 0; 19 } 20</pre>	<pre>Enter a number: 45 Factorial without recursion: 0 === Code Execution Successful ===</pre>

CONCLUSION: Thus, we have successfully executed factorial of a number with and without recursion.

FAQs:

1. What is the purpose of calculating factorial in programming?
2. What is the difference between recursive and non-recursive (iterative) methods?